

# Nikhil Reddy Billa

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## FOCUS AND STRENGTH

A results-driven Machine Learning Engineer specializing in Trustworthy and Responsible AI. My expertise is in designing, building, and deploying end-to-end ML systems that are not only scalable and efficient but also secure, robust, and reliable. My background in AI research, particularly in LLM safety and adversarial robustness, is complemented by hands-on MLOps experience, enabling me to translate cutting-edge research into production-ready solutions. Eager to contribute to a team building impactful AI Product or cutting-edge research.

## SKILLS

**AI/ML Frameworks and Tools:** PyTorch, vLLM, fastai, ML System Design, LLM Fine-tuning (LoRA/QLoRA), RLHF/GRPO, RAG Architectures, Agentic AI (LangGraph, LangChain), scikit-learn, CUDA, ONNX, Hugging Face, DeepSpeed, TensorRT, Model Quantization (bitsandbytes, GGUF), LlamaIndex, MLOps, Inference Optimization

**Programming Languages:** Python, C++, SQL, MATLAB, Bash, C#

**Cloud and DevOps Technologies:** Kubernetes, CI/CD, Docker, Azure (AKS, Functions, Monitor, DevOps, DataFactory, Event Hubs), AWS, AWS bedrock, AWS Codeguru, Pinecone (Vector DB), MLflow

**Data Management and Analysis:** SQL, NoSQL, Data Analysis, Pandas, NumPy, Statistical Modeling, Data Visualization, Spark, Hadoop, Map Reduce

**Software Development Practices:** Agile Development, Version Control (Git), Object-Oriented Programming (OOP), Design Patterns, APIs, Data Structures, Algorithms, Json

**Security & Ethical AI:** Red Teaming, Jail-breaking, Vulnerability Detection, Adversarial Training, Privacy-Preserving Techniques, Secure Code Generation, Extractable Memorization.

## PUBLICATIONS & REVIEWING

- Adversarial Déjà Vu: Jailbreak Dictionary Learning for Stronger Generalization ( ICLR 2026 )
- Retracing the Past: LLMs Emit Training Data When They Get Lost”, (EMNLP Main 2025)
- Data is All You Need (Almost): Iterative Synthetic Instruction Tuning for Secure Code Generation”, (Hokie-Tokie Team), Nova-AI-Challenge proceedings
- ”IDDAW: A Benchmark for Safe and Robust Segmentation of Drive Scenes in Unstructured Traffic and Adverse Weather”, IEEE/CVF Winter Conference on Applications of Computer Vision (WACV) 2024.
- CNN based Image Resizing Forensics for Double Compressed JPEG Images”, Journal of Information Security and Applications, Volume 81, 2024.
- International Multi-centre Validation of Unsupervised Domain Adaptation for Precise Discrimination between Normal and Abnormal Retinal Development, Investigative Ophthalmology & Visual Science.”
- Reviewer of Journal of Information Security and Applications (Elsevier), 02/2025

## WORK EXPERIENCE

Graduate Research Assistant | REDS Lab, Virginia Tech

10/2024

Advisor :Dr.Rouxi Jia

- Architected an end-to-end alignment pipeline for the Amazon Nova Trusted AI Challenge using LoRA/QLoRA and Group Relative Policy Optimization (GRPO) to optimize the Pareto frontier between code utility and 10+ CWE security categories.
- Developed Adversarial Déjà Vu (ICLR 2026), a defense framework leveraging Sparse Dictionary Learning to decompose complex jailbreaks into fundamental adversarial primitives, significantly increasing robustness against unseen multi-turn attack families.
- Engineered the Confusion-Inducing Attack (CIA) framework (EMNLP 2025) to quantify LLM training data leakage; achieved a **22.2% verbatim extraction rate** on Llama-2 (70B) by identifying spikes in **token-level prediction entropy**.
- Built and containerized 13+ automated evaluation harnesses integrating CodeQL, Amazon CodeGuru, and Semgrep with LLM-as-a-Judge architectures for high-throughput, production-grade security assessments.
- Automated the deployment of diverse jailbreak vectors (PAIR, GCG, Crescendo) into a unified red-teaming CI/CD pipeline, enabling rapid iterative hardening of model weights for secure deployment.
- Currently developing a causal error-analysis framework for Multi-Agent System (MAS) using OTel structured traces to detect and categorize failure spans by backtracking through the nodes in complex agentic workflows.

Software Engineer - Machine Learning (MLOps) | NCR Corporation - Hyderabad, India

07/2022 – 08/2024

- Architected a production-grade LLM-driven Observability Stack using Llama-2, automating multi-modal log analysis and reducing mean-time-to-detection (MTTD) by **90%** across globally distributed CI/CD pipelines.
- Developed and deployed an Anomaly Detection Engine on system-level logs utilizing transformer-based sequence modeling, preventing 25+ critical outages and achieving a **40% reduction in system downtime**.

- Engineered a high-performance **RAG (Retrieval-Augmented Generation)** assistant using **LangChain** and **Pinecone** (Vector DB), enabling sub-second latency for technical documentation retrieval during DevOps troubleshooting.
- Orchestrated a **Serverless MLOps Infrastructure** on **Azure (AKS)** using **Kubernetes** and **Docker**, implementing automated CI/CD for model re-training and versioning that accelerated release cycles by **75%**.
- Optimized LLM inference throughput for edge-deployment on POS hardware, leveraging **TensorRT** and quantization techniques to reduce memory footprint by 4x without degrading diagnostic accuracy.

Research Assistant | *ML Lab, International Institute of Information Technology Hyderabad, India*02/2022 - 07/2023

Advisor :**Dr.Girish Varma**

- Aimed at improving the **robustness of semantic segmentation** models under **adverse weather conditions** such as rain, fog, and low light. Made a dataset of 5000 NIR-RGB images.
- Designed a novel **safety metric** to evaluate the performance of segmentation models in autonomous driving scenarios, focusing on **safety and reliability** in adverse weather.
- Developed an AI-based system utilizing OCT scans to **grade arrested retinal development**.
- Incorporated data from different centers and OCT manufacturers and ensured the **system’s robustness to device variations** through the implementation of **domain adaptation** techniques.

Computer Vision Intern | *KoiReader Technologies - Bengaluru, India*03/2021 - 06/2021

- Designed a **document classification** system using **LayoutLM**, customizing it for specific business need.
- Preprocessed raw documents and **created a custom dataset** for model training.
- Trained **LayoutLM models on custom datasets**, improving document classification accuracy by leveraging deep learning and NLP techniques.

Research Assistant | *Intelligent Systems Lab, NIT - Rourkela, India*10/2020 - 01/2022

Advisor :**Dr.Manish Okade**

- Introduced a fully end-to-end CNN architecture featuring a **preprocessing layer with 20 high-pass filters** to suppress image content.
- The **proposed network** is fully end-to-end and **does not rely on any hand-crafting**.

EDUCATION

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|---------------------------------------------------------------------------------------------------------------------|--------------------------------------|
| <b>Virginia Polytechnic Institute and State University, Blacksburg</b><br><i>MS in Computer Engineering</i>         | 08/2024 - 05/2026<br><i>3.63/4.0</i> |
| <b>National Institute of Technology (NIT), Rourkela</b><br><i>Bachelor of Technology in Electrical Engineering.</i> | 07/2018 - 06/2022                    |

POSITIONS, ACTIVITIES AND AWARDS

- **Amazon NOVA Trusted AI Challenge 2024** Innovative solutions in **secure and safe code generation**.
- **Technical Lead, ML4E Club**: Led AI-driven initiatives, mentored students in ML research.